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SCHOOL OF ARTIFICIAL INTELLIGENCE

Introduction to Machine Learning with Pytorch

Syllabus

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Overview

Learn to build powerful models in this Machine Learning Nanodegree with PyTorch. Master supervised, unsupervised, and deep learning techniques through projects involving customer segmentation and image classification.

Nanodegree Program | Intermediate | 49 hours | 4.7 (249 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

- Multivariable calculus
- Data wrangling
- Python for data science
- Basic supervised machine learning
- Intermediate Python
- Linear algebra
- Basic descriptive statistics
- Basic calculus
- Basic probability

You will also need to be able to communicate fluently and professionally **in written and spoken English**.

Skills You'll Learn

Supervised Learning

Naive bayes classifiers | Model evaluation | Support vector machines | Decision trees | Convolutional kernels | scikit-learn | Perceptron | Categorical data visualization | Statistical modeling fundamentals | Chart types | Quantitative data visualization | Linear regression | Spam detection | Logistic regression | Professional presentations | Hyperparameter tuning

Introduction to Neural Networks with PyTorch

Gradient descent | AI algorithms in Python | Training neural networks | NumPy | Backpropagation | Overfitting prevention | Deep learning fluency | PyTorch

Unsupervised Learning

Gaussian mixture models | Single linkage clustering | K-means clustering | Dimensionality reduction | Market segmentation | Cluster models | Principal component analysis | Independent component analysis | Dbscan

Courses

01. Introduction to Machine Learning

2 hours

Welcome to Machine learning with Pytorch

L1	Welcome to Machine Learning	Welcome to the Machine Learning Engineer Nanodegree program! Learn about the program structure and the projects you'll work on in this program.
L2	Getting Help	You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content.
L3	Setting Up Your Computer	In this lesson, get your computer set up with Python 3 using Anaconda, as well as setting up a text editor.

02. Supervised Learning

17 hours

From theory to application, this course guides you through supervised learning essentials. Learn to select, implement, and refine models that solve complex classification and regression tasks.

L1	Introduction to Supervised Learning	Before diving into the many algorithms of machine learning, it is important to take a step back and understand the big picture associated with the entire field.
L2	Linear Regression	Linear regression is one of the most fundamental algorithms in machine learning. In this lesson, learn how linear regression works!
L3	Perceptron Algorithm	The perceptron algorithm is an algorithm for classifying data. It is the building block of neural networks.
L4	Decision Trees	Decision trees are a structure for decision-making where each decision leads to a set of consequences or additional decisions.
L5	Naive Bayes	Naive Bayesian Algorithms are powerful tools for creating classifiers for incoming labeled data. Specifically Naive Bayes is frequently used with text data and classification problems.
L6	Support Vector Machines	Support vector machines are a common method used for classification problems. They have been proven effective using what is known as the 'kernel' trick!

L7	Ensemble Methods	Bagging and boosting are two common ensemble methods for combining simple algorithms to make more advanced models that work better than the simple algorithms would on their own.
L8	Model Evaluation Metrics	Learn the main metrics to evaluate models, such as accuracy, precision, recall, and more!
L9	Training and Tuning	Learn the main types of errors that can occur during training, and several methods to deal with them and optimize your machine learning models.
L10	Project: Finding Donors Project	You've covered a wide variety of methods for performing supervised learning -- now it's time to put those into action!

03. Introduction to Neural Networks with PyTorch

17 hours

Learn the fundamentals of neural networks with Python and PyTorch, and then use your new skills to create your own image classifier—an application that will first train a deep learning model on a dataset of images and then use the trained model to classify new images.

L1	Course Introduction	Meet your instructors, get a short overview of what you'll be learning, check your prerequisites, and learn how to use the workspaces and notebooks found throughout the lessons.
L2	Introduction to Neural Networks	In this lesson, Luis will give you solid foundations on deep learning and neural networks. You'll also implement gradient descent and backpropagation in Python right here in the classroom.
L3	Implementing Gradient Descent	Mat will introduce you to a different error function and guide you through implementing gradient descent using numpy matrix multiplication.
L4	Training Neural Networks	Now that you know what neural networks are, in this lesson you will learn several techniques to improve their training.
L5	Deep Learning with PyTorch	Learn how to use PyTorch for building deep learning models.
L6	Project: Create Your Own Image Classifier	In this project, you'll create your own image classifier and then train—and evaluate its performance—using one of the most classic and well-studied computer vision data sets, CIFAR-10.

04. Unsupervised Learning

14 hours

Learn to apply unsupervised learning methods like K-means and Gaussian mixtures to extract value from raw data. Develop skills in feature extraction and cluster validation to enhance data analysis.

L1	Clustering	Clustering is one of the most common methods of unsupervised learning. Here, we'll discuss the K-means clustering algorithm.
L2	Hierarchical and Density-Based Clustering	We continue to look at clustering methods. Here, we'll discuss hierarchical clustering and density-based clustering (DBSCAN).
L3	Gaussian Mixture Models and Cluster Validation	In this lesson, we discuss Gaussian mixture model clustering. We then talk about the cluster analysis process and how to validate clustering results.
L4	Dimensionality Reduction and PCA	Often we need to reduce a large number of features in our data to a smaller, more relevant set. Principal Component Analysis, or PCA, is a method of feature extraction and dimensionality reduction.
L5	Random Projection and ICA	In this lesson, we will look at two other methods for feature extraction and dimensionality reduction: Random Projection and Independent Component Analysis (ICA).
L6	Project: Project: Identify Customer Segments	In this project, you'll apply your unsupervised learning skills to two demographics datasets, to identify segments and clusters in the population, and see how customers of a company map to them.

05. Congratulations!

1 minute

L1	Congratulations!	You've now reached the end of this program!

06. Prerequisite: Python for Data Analysis

12 hours

L1	Why Python Programming	Welcome to Introduction to Python! Here's an overview of the course.
L2	Data Types and Operators	Familiarize yourself with the building blocks of Python! Learn about data types and operators, built-in functions, type conversion, whitespace, and style guidelines.
L3	Control Flow	Build logic into your code with control flow tools! Learn about conditional statements, repeating code with loops and useful built-in functions, and list comprehensions.

L4	Functions	Learn how to use functions to improve and reuse your code! Learn about functions, variable scope, documentation, lambda expressions, iterators, and generators.
L5	Scripting	Set up your own programming environment to write and run Python scripts locally! Learn good scripting practices, interact with different inputs, and discover awesome tools.
L6	NumPy	Learn the basics of NumPy and how to use it to create and manipulate arrays.
L7	Pandas	Learn the basics of Pandas Series and DataFrames and how to use them to load and process data.

07. Prerequisite: SQL for Data Analysis

16 hours

L1	Basic SQL	In this section, you will gain knowledge about SQL basics for working with a single table. You will learn the key commands to filter a table in many different ways.
L2	SQL Joins	In this lesson, you will learn how to combine data from multiple tables together.
L3	SQL Aggregations	In this lesson, you will learn how to aggregate data using SQL functions
L4	SQL Subqueries & Temporary Tables	In this lesson, you will learn about subqueries, a fundamental advanced SQL topic. This lesson will walk you through the appropriate applications of subqueries, the different types of subqueries, and review subquery syntax and examples.
L5	SQL Data Cleaning	Cleaning data is an important part of the data analysis process. You will be learning how to perform data cleaning using SQL in this lesson.
L6	SQL Window Functions	Window functions allow users to compare one row to another without doing any joins using one of the most powerful concepts in SQL data analysis.
L7	SQL Advanced JOINS & Performance Tuning	Learn advanced joins and how to make queries that run quickly across giant datasets. Most of the examples in the lesson involve edge cases, some of which come up in interviews.

08. Prerequisite: Command Line Essentials

2 hours

L1	Shell Workshop	The Unix shell is a powerful tool for developers of all sorts. In this lesson, you'll get a quick introduction to the very basics of using it on your own computer.

09. Prerequisite: Git & Github

15 hours

L1	What is Version Control	Version control is an incredibly important part of a professional programmer's life. In this lesson, you'll learn about the benefits of version control and install the version control tool Git!
L2	Create a Git Repo	Now that you've learned the benefits of Version Control and gotten Git installed, it's time you learn how to create a repository.
L3	Review a Repo's History	Knowing how to review an existing Git repository's history of commits is extremely important. You'll learn how to do just that in this lesson.
L4	Add Commits to a Repo	A repository is nothing without commits. In this lesson, you'll learn how to make commits, write descriptive commit messages, and verify the changes you're about to save to the repository.
L5	Tagging, Branching, and Merging	Being able to work on your project in isolation from other changes will multiply your productivity. You'll learn how to do this isolated development with Git's branches.
L6	Undoing Changes	Help! Disaster has struck! You don't have to worry, though, because your project is tracked in version control! You'll learn how to undo and modify changes that have been saved to the repository.
L7	Working with Remotes	You'll learn how to create remote repositories on GitHub and how to get and send changes to the remote repository.
L8	Working on Another Developer's Repository	In this lesson, you'll learn how to fork another developer's project. Collaborating with other developers can be a tricky process, so you'll learn how to contribute to a public project.
L9	Staying in Sync with a Remote Repository	You'll learn how to send suggested changes to another developer by using pull requests. You'll also learn how to use the powerful \`git rebase\` command to squash commits together.

10. Additional Material: Python for Data Visualization

13 hours

L1	Data Visualization in Data Analysis	In this lesson, see the motivations for why data visualization is an important part of the data analysis process and where it fits in.
L2	Design of Visualizations	Learn about elements of visualization design, especially to avoid those elements that can cause a visualization to fail.
L3	Univariate Exploration of Data	In this lesson, you will see how you can use matplotlib and seaborn to produce informative visualizations of single variables.
L4	Bivariate Exploration of Data	In this lesson, build up from your understanding of individual variables and learn how to use matplotlib and seaborn to look at relationships between two variables.
L5	Multivariate Exploration of Data	In this lesson, see how you can use matplotlib and seaborn to visualize relationships and interactions between three or more variables.
L6	Explanatory Visualizations	Previous lessons covered how you could use visualizations to learn about your data. In this lesson, see how to polish up those plots to convey your findings to others!
L7	Visualization Case Study	Put to practice the concepts you've learned about exploratory and explanatory data visualization in this case study on factors that impact diamond prices.

11. Additional Material: Statistics for Data Analysis

30 hours

L1	Descriptive Statistics - Part I	In this lesson, you will learn about data types, measures of center, and the basics of statistical notation.
L2	Descriptive Statistics - Part II	In this lesson, you will learn about measures of spread, shape, and outliers as associated with quantitative data. You will also get a first look at inferential statistics.
L3	Admissions Case Study	Learn to ask the right questions, as you learn about Simpson's Paradox.
L4	Probability	Gain the basics of probability using coins and die.
L5	Binomial Distribution	Learn about one of the most popular distributions in probability - the Binomial Distribution.
L6	Conditional Probability	Not all events are independent. Learn the probability rules for dependent events.
L7	Bayes Rule	Learn one of the most popular rules in all of statistics - Bayes rule.

L8	Python Probability Practice	Take what you have learned in the last lessons and put it to practice in Python.
L9	Normal Distribution Theory	Learn the mathematics behind moving from a coin flip to a normal distribution.
L10	Sampling distributions and the Central Limit Theorem	Learn all about the underpinning of confidence intervals and hypothesis testing - sampling distributions.
L11	Confidence Intervals	Learn how to use sampling distributions and bootstrapping to create a confidence interval for any parameter of interest.
L12	Hypothesis Testing	Learn the necessary skills to create and analyze the results in hypothesis testing.
L13	Case Study: A/B tests	Work through a case study of how A/B testing works for an online education company called Audacity.
L14	Regression	Use python to fit linear regression models, as well as understand how to interpret the results of linear models.
L15	Multiple Linear Regression	Learn to apply multiple linear regression models in python. Learn to interpret the results and understand if your model fits well.
L16	Logistic Regression	Learn to apply logistic regression models in python. Learn to interpret the results and understand if your model fits well.

12. Additional Material: Linear Algebra

4 hours

L1	Introduction	Take a sneak peek into the beautiful world of Linear Algebra and learn why it is such an important mathematical tool.
L2	Vectors	Learn about vectors, the basic building block of Linear Algebra.
L3	Linear Combination	Learn how to scale and add vectors and how to visualize the process.
L4	Linear Transformation and Matrices	What is a linear transformation and how is it directly related to matrices? Learn how to apply the math and visualize the concept.

Meet Your Instructors

Josh Bernhard

Staff Data Scientist

Josh has been sharing his passion for data for over a decade. He's used data science for work ranging from cancer research to process automation. He recently has found a passion for solving data science problems within marketplace companies.

Mat Leonard

Content Developer

Mat is a former physicist, research neuroscientist, and data scientist. He did his PhD and Postdoctoral Fellowship at the University of California, Berkeley.

Andrew Paster

Instructor

Andrew has an engineering degree from Yale, and has used his data science skills to build a jewelry business from the ground up. He has additionally created courses for Udacity's Self-Driving Car Engineer Nanodegree program.

Jennifer Staab

Instructor

Jennifer has a PhD in Computer Science and a Masters in Biostatistics; she was a professor at Florida Polytechnic University. She previously worked at RTI International and United Therapeutics as a statistician and computer scientist.

Dan Romuald Mbanga

Instructor

Dan leads Amazon AI's Business Development efforts for Machine Learning Services. Day to day, he works with customers—from startups to enterprises—to ensure they are successful at building and deploying models on Amazon SageMaker.

Cezanne Camacho

Curriculum Lead

Cezanne is an expert in computer vision with a Masters in Electrical Engineering from Stanford University. As a former researcher in genomics and biomedical imaging, she's applied computer vision and deep learning to medical diagnostic applications.

Sean Carrell

Instructor

Sean Carrell is a former research mathematician specializing in Algebraic Combinatorics. He completed his PhD and Postdoctoral Fellowship at the University of Waterloo, Canada.

Jay Alammar

Instructor

Jay is a software engineer, the founder of Qaym (an Arabic-language review site), and the Investment Principal at STV, a $500 million venture capital fund focused on high-technology startups.

Luis Serrano

ML Engineer

Luis was formerly a Machine Learning Engineer at Google. He holds a PhD in mathematics from the University of Michigan, and a Postdoctoral Fellowship at the University of Quebec at Montreal.

Why Udacity

Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- Gain proven experience
- Retain knowledge longer
- Apply new skills immediately

24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- Get help from subject matter experts
- Gain valuable insights and improve your skills
- Learn industry best practices